Fiscal Year:	FY 2012	**Task Last Updated:**	FY 05/02/2012
PI Name:	Perera, Imara Y Ph.D.		
Project Title:	The Effect of Spaceflight on Transgenic Arabidopsis Plants With Compromised Signaling		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	27695	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	07/01/2010	**End Date:**	06/30/2013
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
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Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-135 ; ISS		
Key Personnel Changes/Previous PI:	Caroline M Smith - Research Associate ; Eric S Land- Research Assistant		
COI Name (Institution):	Brown, Christopher (North Carolina State University) Sederoff, Heike (North Carolina State University)		
Grant/Contract No.:	NNX10AM72G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plants provide a complete and economical means for human life support for long-term space exploration and habitation. However, since the space environment is not optimal for plant growth, an understanding of how plants sense and respond to changes in their environment is of fundamental importance. The phosphoinositide (PI) pathway is highly conserved among eukaryotes and functions in the regulation of a multitude of cellular pathways. The lipid-derived second messenger inositol 1,4,5-trisphosphate (InsP3) increases in response to many different stresses. We have shown that InsP3 levels increase with gravistimulation prior to visible bending in both monocot and dicot systems. We have generated transgenic Arabidopsis plants expressing the mammalian type I inositol polyphosphate 5-phosphatase (InsP 5-ptase), an enzyme that specifically hydrolyzes InsP3 and terminates the signal. The transgenic plants have normal growth and morphology; however, they exhibit altered responses to many environmental stimuli including gravity, drought and cold. While rapid changes in transcript levels occur in wild type Arabidopsis within 5 min of gravistimulation, the expression of several of the fastest responding genes does not change in the InsP 5-ptase roots in response to reorientation. Our hypothesis is that InsP3 is an important second messenger in the sensing and signaling of stimuli (including gravity). The plants with compromised InsP3 signaling therefore, provide a valuable tool for dissecting the role of the InsP3 pathway in plant responses to the microgravity environment encountered on the International Space Station. Our primary aim is to identify the molecular changes that are specifically mediated by InsP3 in the space environment. We will compare transcript and protein profiles of wild type and transgenic plants that are grown in both microgravity and 1g in space as well as on the ground. The long term goal is to understand the molecular mechanisms plants use to sense and respond to changes in their environment. This knowledge will help design plants which are better able to withstand space flight and microgravity conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The InsP 5-ptase transgenic plants provide a good model system for understanding several issues including plant responses to gravity, plant responses to environmental stress, and fundamental aspects of PI signaling. Our previous work has implicated InsP3 as a key player in gravity sensing and signaling and the transgenic plants have attenuated InsP3 signaling and reduced gravitropism. In addition these plants have altered responses to different environmental stresses. Therefore this study will help us understand plant signaling during environmental stress, which has long term implications with improving crops to better withstand unfavorable conditions. Furthermore, the phosphoinositide pathway is highly conserved among plants, yeast, and mammals. These studies using the transgenic plants with attenuated InsP3 signaling will provide valuable insights on the regulation and downstream effectors of this pathway and could also contribute to our understanding of the role of the PI pathway in regulating human physiology.
Task Progress:	The major part of last year was spent with preparing the flight build and with the execution of the experiment on board the ISS. 1. The research associate spent 10 days at NASA ARC to participate in the flight build. During this time, she sterilized the wild type and transgenic seeds, mounted the seeds on the membranes and assembled the seed cassettes. (June 2011) 2. The Plant Signaling Experiments were carried to the ISS onboard the final shuttle flight STS-135. The PI and PI research team were fortunate to be able visit Florida and witness the launch. (July 2011) 3. The 2 experimental runs were conducted onboard the ISS during July-Aug 2011. The PI visited NASA ARC and was present for the first experimental run. This included monitoring the start of the experiment (by hydration of the seed cassettes) and monitoring images of seedling growth downloaded every 6hs. The PI was present to witness the end of experimental run 1, EC disassembly by the astronaut and transfer to the MELFI freezer for storage. Experimental runs 1 and 2 were both carried out without any issues. Germination rates and seedling growth were very good in both experimental runs. The samples were stored safely in the freezer and are awaiting return back to Earth. 4. The PI and research associate attended the ASGSB meeting in CA in November 2011 and presented a poster describing the Plant Signaling experiments. 5. A ground control experiment is scheduled to be conducted at NASA ARC in May/June 2012 which will closely mimic the environmental parameters (temperature, CO2) that were encountered during Plant Signaling on orbit runs. The samples from this test will be sent to the PI for analysis. 6. It is anticipated that the flight samples will be returned to Earth on the SpaceX Dragon vehicle and is scheduled for fall of 2012.
Bibliography Type:	Description: (Last Updated: 05/30/2025)
Abstracts for Journals and Proceedings	Smith CM, Land ES, Brown C, Winter-Sederoff H, Perera IY. "The role of InsP3-mediated signaling in plant responses to microgravity." Student Poster Presentation at 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. Program and abstracts. 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. p. 85. , Nov-2011
Significant Media Coverage	Hand E. "Space-station rendezvous set to spur research push. Article describing the Space X/Dragon upcoming launch ; PI Perera interviewed in the article." Nature. 2012 Apr 25;484(7395):426-7. http://www.nature.com/news/space-station-rendezvous-set-to-spur-research-push-1.10501 , Apr-2012
Significant Media Coverage	Smith CM, Land ES, Perera IY. "Interviews on local TV (WRAL, News 14 Carolina), about shuttle launch to ISS and their onboard experiment." WRAL TV, News 14 Carolina, June 2011., Jun-2011
Significant Media Coverage	Perera IY. "Panel discussion - ISS research. " NASA TV, July 2011., Jul-2011
Significant Media Coverage	Ford D. "Countdown to research. Article about PI's research to be flown on July 8, 2011 Shuttle flight to ISS." NCSU Bulletin. March 10, 2011., Mar-2011

Significant Media Coverage	Fagen A. "Members in the news. News story" ASPB News (Newsletter of the American Society of Plant Biologists, September 2011., Sep-2011